

ROUYN-NORANDA AP, QC, Canada

WMO: 717255

Lat: 48.206N

Lon: 78.836W

Elev: 301

StdP: 97.76

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.9	-29.7	-37.6	0.1	-32.8	-34.4	0.2	-29.4	9.2	-8.5	8.4	-9.3	1.2	240	0.655

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	28.9	19.9	27.1	18.8	25.5	17.9	21.4	26.6	20.3	24.9	19.3	23.6	5.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.7	15.0	23.6	18.7	14.0	22.8	17.6	13.1	21.7	63.8	26.9	59.9	24.9	56.4	23.6	25.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.7	7.7	6.9	-37.7	32.0	2.9	1.9	-39.8	33.4	-41.5	34.5	-43.1	35.5	-45.3	36.9		
(5)			-37.8	23.5	2.9	1.1	-39.9	24.3	-41.6	24.9	-43.2	25.6	-45.3	26.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.4	-15.8	-13.8	-7.1	1.2	9.8	15.4	17.7	16.4	12.2	5.3	-2.6	-10.5
	DBStd	13.06	8.19	7.32	7.44	5.71	5.21	4.53	3.49	3.68	4.72	4.84	5.91	7.33
	HDD10.0	3552	801	667	531	267	69	8	0	1	30	164	378	637
	HDD18.3	5897	1059	900	788	513	268	110	54	84	192	406	628	895
	CDD10.0	788	0	0	1	4	64	169	240	198	95	17	1	0
	CDD18.3	91	0	0	0	0	4	22	36	23	7	0	0	0
	CDH23.3	921	0	0	1	2	68	253	339	194	62	1	0	0
CDH26.7	202	0	0	0	0	16	59	81	37	10	0	0	0	
Wind	WSAvg	3.2	3.1	3.3	3.4	3.6	3.4	3.1	3.0	2.9	3.1	3.4	3.5	3.2
Precipitation	PrecAvg	882	58	50	48	56	70	89	104	96	99	82	77	68
	PrecMax	1029	110	95	85	98	129	142	211	138	148	119	132	119
	PrecMin	790	30	25	9	24	30	37	51	49	50	41	51	35
	PrecStd	61	19	18	21	20	26	28	42	27	27	24	19	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.8	5.7	13.6	21.1	28.8	30.9	31.6	29.9	27.9	22.0	13.7	5.6
		MCWB	4.4	3.5	8.2	12.6	17.8	20.6	21.8	20.6	19.5	15.8	10.8	4.3
	2%	DB	1.2	2.1	8.9	16.2	25.0	28.3	28.9	27.4	25.0	18.5	10.0	2.7
		MCWB	0.7	0.5	4.8	8.6	15.2	19.1	20.1	19.6	18.8	13.8	7.8	1.8
	5%	DB	-0.8	0.0	6.0	13.1	22.5	26.3	26.9	25.5	22.7	15.6	7.5	1.1
		MCWB	-1.4	-1.1	2.9	7.1	13.8	17.6	19.1	18.6	17.3	12.4	6.0	0.4
	10%	DB	-3.1	-2.4	3.4	10.2	19.8	24.2	25.1	23.5	20.4	13.1	5.2	0.0
		MCWB	-3.7	-3.4	0.8	5.5	12.6	16.7	18.1	17.7	16.4	10.6	3.7	-0.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.6	4.3	10.5	13.8	20.2	22.4	23.2	22.1	21.2	17.0	10.9	4.8
		MCDB	4.8	5.4	12.0	19.4	25.2	28.9	29.4	27.4	24.8	20.1	12.7	5.3
	2%	WB	0.7	0.7	5.5	9.9	17.2	20.6	21.5	20.8	19.7	14.5	8.3	1.9
		MCDB	1.0	2.0	8.2	13.8	22.6	26.1	26.6	25.5	23.5	17.6	9.5	2.6
	5%	WB	-1.4	-1.2	3.0	7.7	15.3	19.2	20.3	19.6	18.3	12.8	6.0	0.5
		MCDB	-0.8	0.0	5.6	12.2	19.9	24.0	24.8	23.8	21.9	15.3	7.4	1.1
	10%	WB	-3.8	-3.4	0.9	5.7	13.3	17.9	19.2	18.5	16.7	10.7	3.8	-0.6
		MCDB	-3.1	-2.4	3.2	9.8	18.7	22.1	23.5	22.4	20.2	13.0	5.0	-0.1
Mean Daily Temperature Range	5% DB	MDBR	10.8	12.0	12.0	11.2	12.6	12.6	11.6	11.2	10.9	8.3	7.1	8.7
		MCDBR	9.2	10.4	12.9	16.0	16.8	15.4	14.0	13.4	13.4	12.6	9.6	6.7
		MCWBR	9.0	9.3	9.4	10.1	8.7	7.6	7.0	6.9	7.8	8.7	8.1	6.5
	5% WB	MCDBR	9.2	10.4	11.6	14.2	13.8	12.3	11.6	11.4	11.4	11.3	8.8	6.6
		MCWBR	9.1	9.5	8.8	9.8	8.3	7.2	6.3	6.4	7.3	8.6	8.0	6.5
Clear-Sky Solar Irradiance	taub	0.262	0.279	0.313	0.352	0.362	0.383	0.392	0.383	0.366	0.325	0.300	0.262	
	taud	2.262	2.219	2.203	2.305	2.398	2.355	2.361	2.383	2.409	2.511	2.445	2.325	
	Ebn at Noon	811	877	900	898	900	879	864	858	837	825	767	765	
	Edh at Noon	74	98	118	119	113	119	117	110	97	75	64	62	
All-Sky Solar Radiation	RadAvg	1.23	2.32	3.74	4.62	5.21	5.67	5.44	4.57	3.42	2.01	1.04	0.84	
	RadStd	0.19	0.33	0.39	0.39	0.46	0.46	0.36	0.46	0.31	0.32	0.28	0.12	0.11

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page